

Payment Platform Requirements, Design & Use Cases

The settled architecture after three rounds of review: four concepts, one API surface, an extended collection aggregate, and a build sequenced against franchise rather than a scorecard.

Status

For review and implementation

Supersedes

Whitepaper v1.0 & review responses

Requirements

100, with stable identifiers

Use cases

57, mapped to requirements

Contents

1. Scope and decisions taken	3
1.2 Decision log	3
1.3 Out of scope	4
2. Requirements	4
2.1 Commerce — R-CM	4
2.2 Claim — R-CL	5
2.3 Payment — R-PY	5
2.4 Interaction — R-IX	6
2.5 Decision — R-DE	7
2.6 Money — R-MV	7
2.7 Non-functional — R-NF	8
3. Design	8
3.1 The four concepts	8
3.2 Object model and identifiers	9
3.3 Rail abstraction	9
3.4 Decision path	10
4. State model	10
5. Integration model	11
6. Use cases	13
7. Implementation sequence	16
8. Open decisions	17

Scope and decisions taken

This document consolidates the architecture after three review cycles. Positions previously contested are recorded here as settled, with the reasoning compressed to one line each.

1.1 Purpose

A payment platform serving nine channels — enterprise e-commerce, SMB, omni-channel retail, card-present, marketplaces, franchise, platforms and ISVs, PayFacs, and agentic commerce — on one resource model, without a separate stack per segment.

The platform extends the existing Online Payments API rather than replacing it. Merchants integrated today are unaffected and no migration is required.

1.2 Decision log

DECISION	RESOLUTION	REASONING
Number of core concepts	Four, plus Claim	Commerce, Payment, Interaction, Money, with Obligation as a distinct claim. A fifth concept is the signal to re-examine, not to extend.
API versioning	Extend, do not fork	A parallel <code>/v2</code> makes <code>payment</code> mean an attempt and an aggregate simultaneously and permanently. Forked surfaces do not converge.
Collection aggregate name	<code>paymentRequest</code>	Already the production name for this exact object. Developer ergonomics are served by <code>POST /payments</code> being verb-shaped.
Surface binding name	Typed <code>Session</code>	<code>CheckoutSession</code> , <code>TerminalSession</code> , <code>AgentSession</code> , <code>PaymentLink</code> . Resolves the collision with the production <code>paymentRequest</code> .
Order resource exposure	Only for clients without an OMS	A client with an order system never creates a second order. Correlation by <code>order_ref</code> , ledger as a derived read model.
Owed-versus-collected ledger	Derived, never written	No POST, no line items, no synchronisation. Prevents the dual-write failure mode outright.
Obligation	First-class, shipped thin	Invisible on the simple path. Required by marketplace, franchise, PayFac and tax remittance; retrofit cost is disproportionate.
Rail modelling	Rail-specific state beneath one Payment	Card semantics must not leak into the aggregate. RTP and Request-to-Pay are instances of a rule, not exceptions.
Screening	Constraint, never an input to optimisation	Compliance defines the feasible space. The optimiser never sees an excluded option.
Routing objective	Cost-aware, observable terms only	Lifetime value is not observable at authorisation. An objective with unobservable terms cannot pass model validation.
Competitive posture	Lead 4, parity 8, behind 4	Not parity on every dimension. The four "behind" items need volume, time or hardware, not engineering.
First segment	Franchise	Underserved, incumbent is a competitor rather than a native, and it requires no new licensing.

1.3 Explicitly out of scope for v1

- **Simulation and counterfactual APIs.** Require transaction volume we do not have. Building the measurement loop before there is anything to measure is premature.
- **Long-horizon fund holding.** Gated on a compliance and treasury position that is not yet established. Booking and rental marketplaces depend on it and are therefore also deferred.
- **Acquiring licences.** Named as the principal blocker on authorisation rate, but a corporate action rather than a platform deliverable. Tracked separately.
- **Hardware.** Certify and support a partner fleet. We do not manufacture terminals.

SECTION 2

Requirements

100 requirements with stable identifiers. **MUST** is required for v1 acceptance; **SHOULD** is required for segment readiness; **MAY** is deferred but must not be designed out.

2.1 Commerce — R-CM

ID	LEVEL	REQUIREMENT
R-CM-01	MUST	An Order may exist with zero payment objects, and may be cancelled or closed without any payment ever being created.
R-CM-02	MUST	Order lifecycle transitions are driven exclusively by commercial events. No authorization, capture or settlement event may change Order fulfilment state.
R-CM-03	MUST	Order financial state is derived from money objects and is never directly settable by a client.
R-CM-04	MUST	A client that operates its own order management system is never required to create an Order resource. <code>order_ref</code> alone must be sufficient.
R-CM-05	MUST	A client may declare <code>order_total</code> on a payment without creating an Order resource, and read a derived ledger keyed on <code>order_ref</code> .
R-CM-06	MUST	The derived ledger exposes owed, collected, refunded and open amounts, plus tender lines. It has no write endpoint.
R-CM-07	MUST	Externally-settled tenders (third-party gift card, loyalty, store credit) may be registered so that <code>open</code> is correct.
R-CM-08	MUST	An Order supports N payment objects. Split tender, deposit-plus-balance and multi-rail collection are ordinary, not exceptional.
R-CM-09	MUST	An amendment changing amount owed after authorization emits a reconciliation event, resolved by policy to exactly one of: incremental authorization, partial reversal, supplemental payment, or reauthorization.
R-CM-10	MUST	The platform never captures an amount different from the amount authorized without an explicit, policy-permitted overcapture allowance.
R-CM-11	SHOULD	Order supports fulfilment modes of shipment, pickup, delivery and in-store, with per-line fulfilled and returned quantities.
R-CM-12	SHOULD	Order-level refund resolution supports declared policies: proportional, original-tender-first, and explicit.
R-CM-13	SHOULD	Exchange semantics distinguish value returned in exchange from money actually refunded.
R-CM-14	MAY	Order carries a quote state permitting conversion to an open order without re-creation.

2.2 Claim — R-CL

ID	LEVEL	REQUIREMENT
R-CL-01	MUST	An Obligation is a stateful claim held by a party against an order or a period.
R-CL-02	MUST	Party type is polymorphic: connected account, platform, franchisor, tax authority, or nested party. It is not constrained to sub-merchants.
R-CL-03	MUST	Obligation carries a declared <code>basis</code> : line items, percentage of gross, percentage of net, percentage of settled, fixed, declared tax, or contract-defined.
R-CL-04	MUST	Obligation carries a <code>source</code> : order, period, external declaration, or standalone. Period-sourced obligations have no order.
R-CL-05	MUST	The single-obligation case is created implicitly and never surfaces on the simple integration path.
R-CL-06	MUST	The sum of obligations against an order never exceeds amount owed, evaluated at every valuation rather than only at creation.
R-CL-07	MUST	Line items carry a beneficiary grouping key from day one, whether or not multi-party features are in use.
R-CL-08	SHOULD	Obligations accept externally-declared revenue with attestation, so gross-basis claims can span revenue the platform never processed.
R-CL-09	SHOULD	Each obligation carries an independent release condition and clock, unaffected by order state or by other obligations.
R-CL-10	SHOULD	A refund landing after disbursement creates a clawback that nets against future obligations for the same party per a declared netting policy.
R-CL-11	SHOULD	A full valuation audit trail is retrievable: gross, each deduction, and each linked obligation.
R-CL-12	MAY	Obligations support a valuation policy deriving amount from elapsed time, for cancellation tiers and milestone release.
R-CL-13	MAY	Obligations support adjudication for claims contested between two parties who are both customers.

2.3 Payment — R-PY

ID	LEVEL	REQUIREMENT
R-PY-01	MUST	The collection aggregate is creatable before the first authorization message is emitted.
R-PY-02	MUST	Idempotency is mandatory on every mutating request and is scoped to the aggregate and an attempt key. Requests without a key are rejected.
R-PY-03	MUST	A replayed idempotency key with an identical body returns the original response; with a divergent body it is rejected rather than silently served.
R-PY-04	MUST	Idempotency is never keyed on <code>order_ref</code> .
R-PY-05	MUST	The aggregate records every attempt including declines, with network response data, acquirer selected, and routing rationale.
R-PY-06	MUST	A soft decline followed by a successful retry resolves to one aggregate with a successful outcome, not two unrelated objects.
R-PY-07	MUST	Terminal failure requires attempt-ladder exhaustion or a scheme do-not-retry signal. A single soft decline never produces terminal failure.
R-PY-08	MUST	A resumable state exists for shopper-pending interruptions: 3DS challenge, redirect, Pay-by-Bank approval, Request-to-Pay acceptance, and human-in-the-loop confirmation.
R-PY-09	MUST	Authorization and capture are distinct exposed states. No single state may be ambiguous as to whether money moved.
R-PY-10	MUST	Refunds are children of captures. A fully refunded aggregate remains in a captured state.

ID	LEVEL	REQUIREMENT
R-PY-11	MUST	Multiple partial captures are supported against one authorization, each carrying its own Level 3 data.
R-PY-12	MUST	Stored-credential fields are derived from a mandate, not accepted from the client: initiator type, stored-credential status, sequence, purpose, and the network transaction ID chain.
R-PY-13	MUST	Authorization expiry is monitored against the fulfilment horizon and warned before it lapses.
R-PY-14	MUST	Reauthorization creates a new attempt under the same aggregate, chained to the original network transaction ID.
R-PY-15	MUST	Money is represented as integer minor units paired with an ISO 4217 code. No floating-point amount appears in any request, response, webhook or export.
R-PY-16	SHOULD	Incremental authorization and overcapture are supported where scheme and BIN permit, with an explicit computed ceiling.
R-PY-17	SHOULD	Partial authorization is supported, returning the shortfall and remaining open amount so a second tender can be prompted.
R-PY-18	SHOULD	The vault token is the sole element crossing the cardholder-data-environment boundary, and resolves identically across channels.
R-PY-19	SHOULD	Network tokens are requested by default for card-on-file and recurring, with reporting of whether one was used per attempt.
R-PY-20	MAY	Credential selection considers token health, issuer behaviour, cost and approval propensity.

2.4 Interaction — R-IX

ID	LEVEL	REQUIREMENT
R-IX-01	MUST	A Session binds one collection aggregate to one presentation surface and holds no money state.
R-IX-02	MUST	A Session is optional. A client operating its own checkout UI confirms directly against the aggregate and never creates one.
R-IX-03	MUST	Session completion does not imply payment completion. A completed session with an authorized-not-captured aggregate is valid.
R-IX-04	MUST	Only one Session may be in progress for a given aggregate at a time. A concurrent binding is rejected.
R-IX-05	MUST	An aggregate supports N sequential Sessions. A timed-out terminal session can be replaced by a payment link against the same aggregate with no duplicate-charge exposure.
R-IX-06	MUST	Session expiry behaviour is declared: cancel the aggregate, leave it open, or rebind to a named surface type.
R-IX-07	SHOULD	Supported surface types: web SDK, native SDK, hosted checkout, payment link, invoice, terminal, agent, Request-to-Pay, and virtual terminal.
R-IX-08	SHOULD	Method availability is returned per session with a reason when a method is unavailable.
R-IX-09	SHOULD	Card-present is served by a device platform, not by a session alone: device identity, provisioning, key injection, EMV parameters, remote configuration, offline controls and device health.
R-IX-10	SHOULD	Offline acceptance enforces four ceilings — per transaction, per device cumulative, per card cumulative, and maximum offline window — with EMV-only eligibility and FIFO auto-forward.
R-IX-11	SHOULD	Live uncommitted offline exposure is retrievable per device.
R-IX-12	MUST	An agent never receives a reusable credential. Agent-initiated payments use a single-use token bound to a cart hash, a merchant, a ceiling and a short expiry.
R-IX-13	MUST	Agent purchases validate agent identity, delegation grant validity, spend caps, merchant policy, category policy and merchant-side agent policy before any credential is minted.
R-IX-14	SHOULD	Human-in-the-loop confirmation is triggerable by amount, new merchant, or recurring mandate establishment, with a declared timeout behaviour.
R-IX-15	SHOULD	Externally-issued agent mandates are accepted as authorization evidence, and platform-issued grants are exportable in an interoperable form.

2.5 Decision — R-DE

ID	LEVEL	REQUIREMENT
R-DE-01	MUST	Screening resolves before optimisation and defines the feasible option set. The optimiser is structurally incapable of selecting an excluded option.
R-DE-02	MUST	Screening covers sanctions, AML, watchlists, merchant, country, network, regulatory and product restrictions.
R-DE-03	MUST	Capability, entitlement, preference and policy are resolved as four separate questions. "Can transact" and "can be paid" are distinct entitlements.
R-DE-04	MUST	Retry decisions are made by the platform under scheme constraints, not by client code. A client cannot exceed a scheme retry ceiling by configuration.
R-DE-05	MUST	Network advice codes mandating no further attempts terminate the ladder regardless of client instruction.
R-DE-06	MUST	Risk assessment is returned synchronously on the payment object, with score, recommendation, rules fired and signals.
R-DE-07	MUST	Every automated decision records the propensity with which it was taken, enabling off-policy evaluation without live experimentation.
R-DE-08	MUST	Every routing, authentication and retry decision carries a machine-readable explanation with the policy and model versions applied.
R-DE-09	SHOULD	Merchant configuration is compiled into a signed, versioned plan bundle. The authorization path looks up a plan rather than querying policy services per transaction.
R-DE-10	SHOULD	The routing objective uses only terms observable at request time or estimable from observed outcomes: approval propensity, transaction value, processing cost, expected interchange, expected fraud loss, retry cost and FX cost.
R-DE-11	SHOULD	Authentication strategy is coordinated with risk: exemption eligibility, exemption request, soft-decline handling and step-up form one decision rather than two services.
R-DE-12	SHOULD	Models are versioned with documented lineage and challenger evaluation, aligned to supervisory model-risk expectations.
R-DE-13	MAY	Counterfactual estimation answers what would have happened under an alternative route.

2.6 Money — R-MV

ID	LEVEL	REQUIREMENT
R-MV-01	MUST	All money movement produces balanced double-entry ledger transactions. Balances are derived, never stored as mutable fields.
R-MV-02	MUST	Payment status is not treated as financial truth. The ledger is the financial record.
R-MV-03	MUST	Every money row carries the join keys for order reference, aggregate, obligation and ledger transaction at write time.
R-MV-04	MUST	Balance components are separately visible: available, pending, reserved, obligated and negative.
R-MV-05	MUST	Negative balance recovery follows a declared order and emits an event at each step.
R-MV-06	SHOULD	Six currency roles are modelled and auditable end to end: merchant, transaction, settlement, funding, issuer and acquirer.
R-MV-07	SHOULD	FX quotes disclose markup in basis points against a stated mid-market rate, with source, timestamp and lock duration.
R-MV-08	SHOULD	Rolling and risk-triggered reserves are visible to the sub-merchant as a distinct balance component, never as an unexplained shortfall.
R-MV-09	SHOULD	Disputes are part of the payment lifecycle, with evidence assembled from order, authentication, device, session, agent, fulfilment and communication records.

ID	LEVEL	REQUIREMENT
R-MV-10	SHOULD	Reports for a closed period are immutable and byte-reproducible. Late adjustments appear in the following period with an explicit reference, never as a silent restatement.
R-MV-11	SHOULD	Beneficiary reporting for tax purposes is generated from persisted obligation records rather than reconstructed splits.
R-MV-12	MAY	Scheduled collection from beneficiary accounts supports contract-defined periodic obligations.

2.7 Non-functional — R-NF

ID	LEVEL	REQUIREMENT
R-NF-01	MUST	Authorization path availability of 99.99% measured monthly, with no scheduled maintenance window on that path.
R-NF-02	MUST	Recovery point objective of zero for money-bearing writes. Money-bearing reads never served from a lagging replica.
R-NF-03	MUST	Issuer round-trip time reported separately from platform latency, so merchants can distinguish the two.
R-NF-04	MUST	PCI DSS Level 1 service provider, with CDE segmentation validated by penetration test at least semi-annually.
R-NF-05	MUST	Every state transition emits an immutable event. The event log is the audit trail; the object is a projection.
R-NF-06	MUST	Webhook delivery is at-least-once with HMAC signature, timestamp-bound replay protection and a documented retry ladder.
R-NF-07	MUST	Breaking changes ship only as new dated versions. Webhook payloads render at the version pinned on the endpoint, not the caller.
R-NF-08	SHOULD	Authorization latency targets of 180ms at p50, 420ms at p95 and 900ms at p99, measured edge to edge excluding issuer time.
R-NF-09	SHOULD	Sustained throughput of 15,000 transactions per second with peak headroom of at least three times seasonal maximum.
R-NF-10	SHOULD	Cardholder data pinned to region of acquisition. Cross-region access proxied, never replicated.
R-NF-11	SHOULD	Vault export to a competing processor available on request, without fee or delay.
R-NF-12	SHOULD	Time to first successful test payment under fifteen minutes, requiring at most one API call and no new concepts.
R-NF-13	SHOULD	Deterministic time simulation for all time-dependent behaviour, so renewal, proration and dunning defects are reproducible before production.

SECTION 3

Design

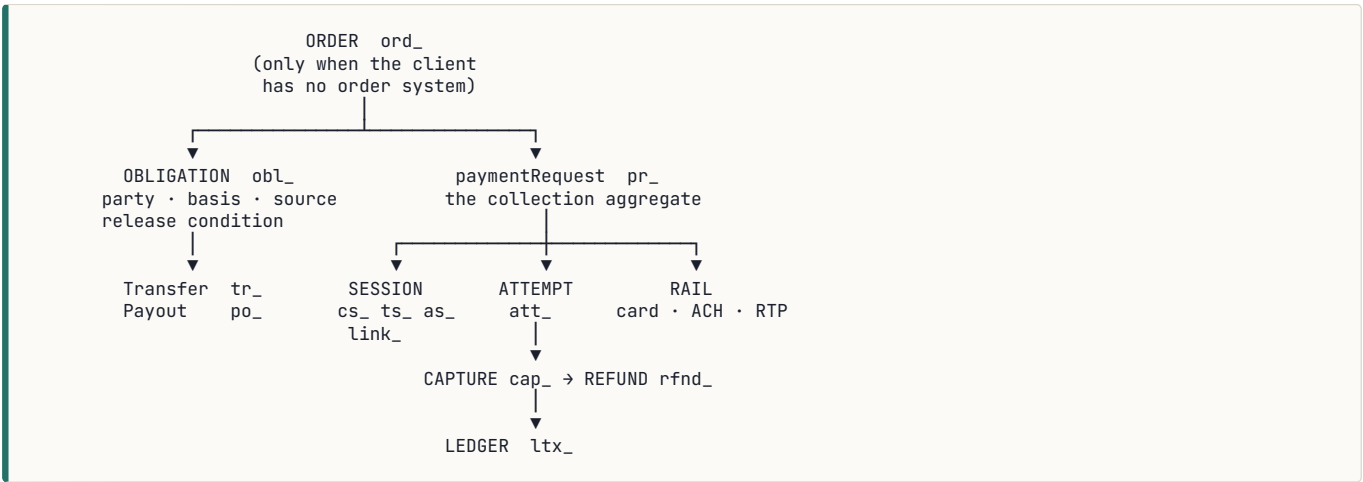
Four concepts, one claim object, one API surface. Everything sophisticated sits behind these and nothing introduces a fifth.

3.1 The four concepts

COMMERCE	What is being bought?	Order · LineItem · Fulfilment
CLAIM	Who has an economic claim?	Obligation
PAYMENT	How much are we collecting?	paymentRequest · Attempt · Capture · Refund
INTERACTION	How does the customer engage?	Session — web · terminal · link · agent
MONEY	Where did the money actually go?	Ledger · Settlement · FX · Payout

Claim is kept distinct from Money rather than folded into it. An obligation can exist, revalue and be disputed long before any money moves, and a clawback is a claim that survives after the movement completed. Collapsing the two loses both cases.

3.2 Object model and identifiers



Identifier registry

PREFIX	RESOURCE	PREFIX	RESOURCE	PREFIX	RESOURCE
ord_	Order	pr_	paymentRequest	cs_	CheckoutSession
obl_	Obligation	att_	Attempt	ts_	TerminalSession
cust_	Customer	cap_	Capture	as_	AgentSession
ca_	ConnectedAccount	rfnd_	Refund	link_	PaymentLink
vt_	VaultToken	dsp_	Dispute	dev_	Device
mnd_	Mandate	ltx_	LedgerTransaction	loc_	Location
agt_	Agent	tr_	Transfer	fxq_	FXQuote
grant_	DelegationGrant	po_	Payout	rsv_	Reserve
brnd_	Brand	roy_	RoyaltySchedule	evt_	Event

3.3 Rail abstraction

The aggregate exposes one lifecycle. Each rail owns its own state machine beneath it. Rail-specific states never leak upward.

paymentRequest (rail-agnostic)

— Card

— ACH

— RTP / FedNow

— Open banking

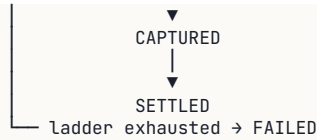
— Request-to-Pay

— AFT / OCT

authorize → capture → clear → settle
initiate → accept → return/reject → settle
initiate → confirm → settle (no capture state)
initiate → consent → settle
issue → deliver → accept → funded → reconciled
create transfer → execute → confirm

Rule: the aggregate must not assume card semantics. Rails without a capture step move from awaiting-action directly to captured on funds receipt.

Rules:



4.3 Session

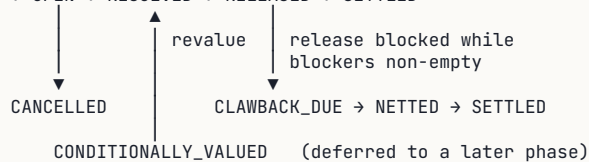
CREATED → ACTIVE → IN_PROGRESS → COMPLETED



Golden rule: session completion never determines payment completion.
Session COMPLETED with aggregate AUTHORIZED is a normal pairing.

4.4 Obligation

CREATED → OPEN → RESOLVED → RELEASED → SETTLED



4.5 Operation propagation

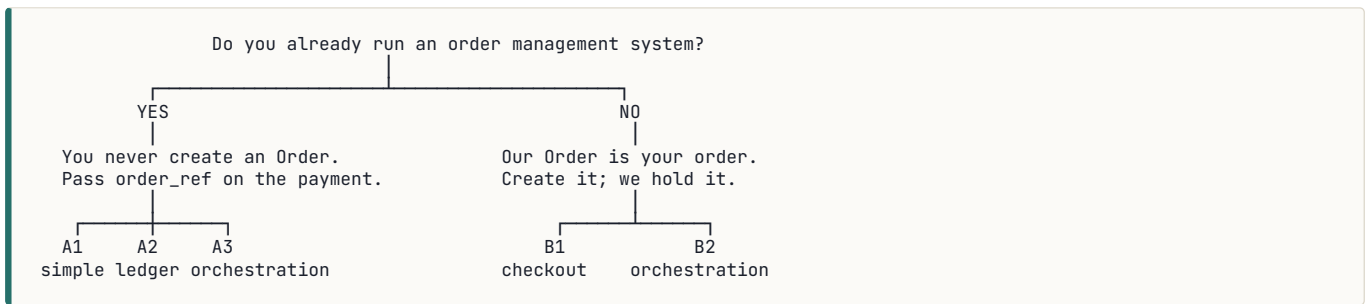
OPERATION	PAYMENTREQUEST	ORDER	OBLIGATION
Authorize	AUTHORIZED	uncollected	created, open
Capture	PARTIALLY_CAPTURED then CAPTURED	partly collected then collected	valued, then payable
Void	CANCELLED	back to uncollected	cancelled
Refund	unchanged: CAPTURED	refund due	revalued, or clawback if disbursed
Dispute	unchanged: CAPTURED	unchanged	disputed, release held

Order and Obligation states are derived. No operation sets them directly.

SECTION 5

Integration model

The first question is not how sophisticated a client is. It is whether they already own an order.



5.1 Branch A — the client owns the order

```
A1 POST /payments                                ships today, unchanged
{ "amount": 10000, "currency": "USD", "captureMethod": "NOW",
  "merchantOrderNumber": "WEB-2026-0884213",
  "paymentMethodType": { "card": { ... } } }

A2 POST /payments                                declare a total, read a derived ledger
{ "order_ref": "WEB-2026-0884213", "order_total": 32200,
  "amount": 19000, ... }

GET /ledger?order_ref=WEB-2026-0884213
→ { "owed": 32200, "collected": 19000, "refunded": 0, "open": 13200,
    "tenders": [ { "type": "card", "amount": 19000, "payment": "pr_9f18..." } ] }

A3 POST /payments/requests                        creatable pre-authorization
POST /payments/requests/{id}/sessions
POST /payments/requests/{id}/confirm
POST /payments/requests/{id}/captures
```

5.2 Branch B — the client has no order system

```
B1 POST /orders                                  → ord_1Pm4KqYd8vRnT2Bx
POST /orders/{id}/checkout                      → one call wires everything

→ { "order": "ord_1Pm4KqYd8vRnT2Bx", "payment": "pr_9f186d77",
    "session": "cs_8Kq2WdNm5Vp",
    "checkout_url": "https://pay.../c/cs_8Kq2WdNm5Vp",
    "payment_status": "requires_instrument" }

B2 As B1, plus explicit orchestration identical to A3.
```

THE RULE THAT PREVENTS THE PRINCIPAL FAILURE MODE

`/ledger` is a read model keyed on `order_ref`. It has no write endpoint, holds no line items, and must never be synchronised. This must be stated prominently in the documentation — *this is not a copy of your order; do not write a sync job* — because that sentence prevents more integration failures than any schema decision.

5.3 What is identical on every path

- The same objects exist. An A1 payment still creates an aggregate and an attempt, created implicitly and returned rather than requested.
- One state machine, one transition set, one event catalogue regardless of entry point.
- The same reconciliation keys, so finance sees no difference between an A1 and a B2 client.
- Moving deeper is additive. No client ever crosses branches, and nothing built is discarded.

Use cases

Each case names the branch it uses and the requirements it exercises. Requirement coverage is the acceptance basis.

6.1 Enterprise e-commerce — ENT

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
ENT-01	Authorize at checkout, capture at ship	A1	R-PY-09, R-PY-11, R-CM-02	Manual capture; fulfilment event triggers capture with Level 3 data.
ENT-02	Multi-node ship from three sites	A1	R-PY-11, R-CM-11	One aggregate, N captures, each with its own Level 3 payload.
ENT-03	Item added after authorization	A2	R-CM-09, R-PY-16	Amendment emits reconciliation; policy resolves to increment or supplemental.
ENT-04	Backorder outlives the authorization	A1	R-PY-13, R-PY-14	Expiry warning, then reauthorize chained to the original network transaction ID.
ENT-05	Retry after soft decline, no duplicate	A1	R-PY-02, R-PY-06, R-DE-04	New attempt under the same aggregate; ladder bounded by scheme policy.
ENT-06	Split tender: card, gift card, loyalty	A2	R-CM-07, R-CM-08, R-CM-05	Two aggregates plus one externally-settled tender line; ledger drives the prompt.
ENT-07	Finance asks what an order collected	A2	R-CM-06, R-MV-03	Derived ledger; join keys present at write time.
ENT-08	Chargeback defence with cart evidence	A1	R-MV-09	Frozen commercial-context snapshot populates the evidence package.
ENT-09	Cross-border with disclosed FX	A1	R-MV-06, R-MV-07	Locked quote referenced on the aggregate; markup reported per capture.
ENT-10	Authorization-rate experimentation	A3	R-DE-07, R-DE-09, R-DE-10	Plan bundles are configuration; propensity logging enables off-policy evaluation.

6.2 SMB and hosted checkout — SMB

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
SMB-01	Sell with no website	B1	R-IX-07	Payment link materialises order and aggregate per presentation.
SMB-02	Hosted checkout, minimal PCI scope	B1	R-IX-01, R-NF-12	One call from order to checkout URL.
SMB-03	Same-day onboarding, sell immediately	B1	R-DE-03	Can-transact granted provisionally while can-be-paid remains blocked.
SMB-04	Refund without finding the transaction	B1	R-CM-12	Order-level refund resolution against tender lines.
SMB-05	Understand a decline in plain language	B1	R-DE-08	Merchant-readable message beside the technical code and network detail.
SMB-06	Occasional in-person acceptance	B1	R-IX-09	Tap-to-phone device registered against the same model as hardware.

6.3 Omni-channel retail — OMN

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
OMN-01	Buy online, collect in store	A2	R-CM-03, R-PY-13	Collectibility gate tells the store whether release is safe.
OMN-02	Pickup delayed past expiry	A2	R-PY-13, R-PY-14	Automatic reauthorization under policy, invisible to the associate.
OMN-03	Return in store, bought online	A2	R-PY-18, R-CM-12	Same vault token across channels resolves the original tender.
OMN-04	Return without a receipt	A2	R-PY-18	Reverse resolution from instrument to order.
OMN-05	Exchange across price points	A2	R-CM-13	Refund against original capture plus a new aggregate; both legs recorded.
OMN-06	Partial return, split tender and loyalty	A2	R-CM-07, R-CM-12	Proportional allocation across tender lines with visible arithmetic.
OMN-07	Terminal times out, shopper pays on phone	A3	R-IX-04, R-IX-05	New session against the same aggregate; no duplicate-charge exposure.
OMN-08	Pay in store, deliver in six weeks	A3	R-PY-13, R-PY-16	Extended authorization, scheduled reauth, increment for delivery fees.

6.4 Card-present and unattended — CP

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
CP-01	Countertop sale with tip prompt	A3	R-IX-09, R-PY-16	Tip returned on the attempt; overcapture within MCC ceiling.
CP-02	Prepaid card with insufficient balance	A2	R-PY-17, R-CM-08	Partial approval returns shortfall and open amount for a second tender.
CP-03	Unattended kiosk or fuel dispenser	A3	R-IX-09, R-PY-16	Status-check authorization, increment, capture at dispensed amount.
CP-04	Connectivity loss mid-shift	A3	R-IX-10, R-IX-11	Four-ceiling offline policy, EMV-only eligibility, FIFO auto-forward.
CP-05	Least-cost debit routing	A1	R-DE-10, R-DE-08	Network selection recorded with rationale as compliance evidence.
CP-06	Fleet firmware and key management	A3	R-IX-09	Device platform handles injection, EMV parameters and scheduled updates.

6.5 Marketplaces — MKP

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
MKP-01	Multi-seller cart, one buyer charge	B2	R-CL-01, R-CL-03, R-CL-06	One aggregate, N obligations on a line-item basis.
MKP-02	Seller onboarding incomplete at first sale	B2	R-DE-03, R-CL-09	Capture succeeds; release blocked with a stated reason. Buyer unaffected.
MKP-03	Refund after the seller was paid	B2	R-CL-10	Clawback nets against future obligations per declared policy.
MKP-04	Platform fee and tax remittance	B2	R-CL-02, R-CL-03	Tax authority as a party type with no account and no balance.

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
MKP-05	Seller disputes a short payout	B2	R-CL-11	Full valuation trail: gross, each deduction, each linked obligation.
MKP-06	Beneficiary tax reporting at scale	B2	R-MV-11	Generated from persisted obligations, not reconstructed splits.
MKP-07	Per-seller release on that seller's delivery	B2	R-CL-09	Independent release condition and clock per obligation.

6.6 Franchise — FRC

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
FRC-01	Royalty on contract-defined gross sales	A2	R-CL-03, R-CL-04, R-CL-08	Period obligation, contract-defined basis, multiple revenue sources.
FRC-02	Cash and delivery revenue in the base	A2	R-CL-08	Attested external declaration; delivery counted at full order value.
FRC-03	Ad fund at a different rate	A2	R-CL-03	Second component on the same schedule, reported separately.
FRC-04	Gift card sold at A, redeemed at B	A2	R-CL-01, R-CL-02	Claim between two sub-merchants with no payer and no charge to attach to.
FRC-05	Monthly royalty sweep	A2	R-MV-12, R-CL-09	Filing-calendar release; scheduled collection from franchisee accounts.
FRC-06	Declared revenue looks understated	A2	R-CL-08, R-CL-11	Variance detection; reconciliation against deposit data where held.
FRC-07	Consistent descriptors network-wide	A1	R-DE-03	Brand-level template that an individual location cannot override.
FRC-08	Cross-location customer identity	A2	R-PY-18	One vault token across locations; no per-location customer fragmentation.

6.7 Platforms, PayFacs and agentic — PLT · PF · AGT

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
PLT-01	Sophisticated and simple customers, one integration	A/B	R-CM-04, R-CM-05	Branch is per sub-merchant configuration, not per platform.
PLT-02	Sub-account data isolation	B2	R-DE-03	Entitlement enforced on every read, not in application code.
PLT-03	Platform fee by tier	B2	R-CL-03	Fee as an obligation with a tier-driven rate; configuration, not code.
PF-01	Provisional onboarding with payout blocked	B1	R-DE-03	Can-transact and can-be-paid resolved as separate entitlements.
PF-02	Sub-merchant goes negative after chargebacks	B2	R-MV-05, R-CL-10	Declared recovery order with an event at each step.
PF-03	Rolling reserve on a high-risk sub-merchant	B2	R-MV-08	Visible as a distinct balance component, never an unexplained shortfall.

ID	USE CASE	PATH	REQUIREMENTS	MECHANICS
PF-04	Migration in from another provider	A/B	R-NF-11	Bulk vault import without re-collecting credentials from cardholders.
AGT-01	Human-present agent purchase	B2	R-IX-12, R-IX-13	Signed cart mandate; single-use token bound to cart, merchant and ceiling.
AGT-02	Autonomous purchase within a standing grant	B2	R-IX-13, R-IX-14	Full constraint pipeline before any credential is minted.
AGT-03	Amount crosses the confirmation threshold	B2	R-IX-14, R-PY-08	Resumable pending state; declared timeout behaviour.
AGT-04	Agent credential is compromised	B2	R-IX-12	Blast radius is one cart at one merchant under one ceiling.
AGT-05	Externally-issued agent mandate presented	B2	R-IX-15	Accepted as authorization evidence; grants exportable in kind.

SECTION 7

Implementation sequence

Five phases with gates. Only Phase 1 is proposed for approval now.

PHASE	SCOPE	REQUIREMENTS	ACCEPTANCE
1 — Foundation Q1	Obligation thin. Aggregate creatable pre-authorization with attempts. Rail abstraction in the model, card implemented. Derived ledger.	R-CL-01...07, R-PY-01...15, R-CM-01...10	A1 unchanged and regression-clean. A2 ledger correct under split tender. Single-obligation path invisible to merchants.
2 — Decision Q2	Screening as constraint. Configuration compiler and signed plan bundles. Decision explanation. Cost-aware routing.	R-DE-01...10, R-NF-08	Optimiser structurally cannot select a screened-out option. Latency targets met with plan lookup rather than live policy resolution.

PHASE	SCOPE	REQUIREMENTS	ACCEPTANCE
3 — Franchise Q2–Q3	Period obligations. External revenue declaration with attestation. Royalty sweep. Brand-level descriptors and gift-card liability.	R-CL-08, R-CL-11, R-MV-12, R-DE-03	One design partner computing royalty on contract-defined gross including non-processed revenue.
4 — Parity Q3–Q4	Money platform depth, currency roles, dispute lifecycle, card-present device platform, billing engine.	R-MV-06...11, R-IX-09...11, R-NF-13	Sequenced against observed deal loss, not against a capability checklist.
5 — Contingent Deferred	Obligation revaluation and release clocks. Simulation and counterfactual estimation. Long-horizon fund holding.	R-CL-12, R-CL-13, R-DE-13	Gated on compliance position and on volume sufficient to make measurement meaningful.

Gates

GATE	CRITERION, AND THE ACTION IF IT FAILS
Day 90	Design partners report no measurable integration friction on the single-obligation path. <i>If it fails:</i> collapse Obligation to a line-item attribute.
Day 90	At least five franchisor conversations converted to qualified discovery. <i>If it fails:</i> stop Phase 3; keep the thin object.
Day 180	Compliance and treasury position on long-horizon fund holding documented either way. <i>If negative:</i> Phase 5 fund holding removed; franchise unaffected.
Day 270	One franchise design partner committed before Phase 4 investment is released.

SECTION 8

Open decisions

Six items this document does not settle. The first three block Phase 1 scope.

#	QUESTION	WHAT IT DECIDES
1	Can the aggregate identifier be minted at the API layer before authorization, or is it derived downstream?	Blocks R-PY-01. If it cannot be minted independently, the creatable aggregate is unavailable and Branch A3 collapses to A1 and A2.
2	Is the existing request identifier replay-safe? Does resending after a timeout return the original response, reject, or reprocess?	Decides R-PY-03 scope. If replay-safe today, the requirement is documentation rather than implementation.
3	Does a declined authorization currently produce an aggregate?	Decides R-PY-05 scope. If declines are already children, the attempts work reduces to adding routing metadata.
4	Does <code>order_total</code> belong on the first payment only, or on every payment against an order reference?	Likely resolution: accept on any payment, last write wins, emit a reconciliation event on change. Needs a documented multi-tender example before launch.
5	Do we publish Branch A3 and B2 at launch, or hold them behind an advanced section?	Publishing everything risks clients self-selecting into orchestration unnecessarily — the over-complexity the reviews warned about.

#	QUESTION	WHAT IT DECIDES
6	Is a new major version warranted on capability grounds?	Not for naming. Only if rail abstraction, obligations or the state machine prove genuinely un-additive — and then only with a funded migration and a dated sunset agreed in advance.

BASIS FOR REVIEW

Approval sought for **Phase 1 only**, together with the gate schedule in Section 7. Requirements marked MUST in the Commerce, Claim and Payment domains constitute the acceptance basis. Items 1 to 3 above should be resolved with the owning API team before Phase 1 scope is fixed, since each can materially reduce or expand it.